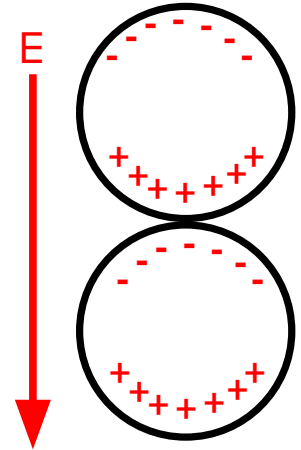
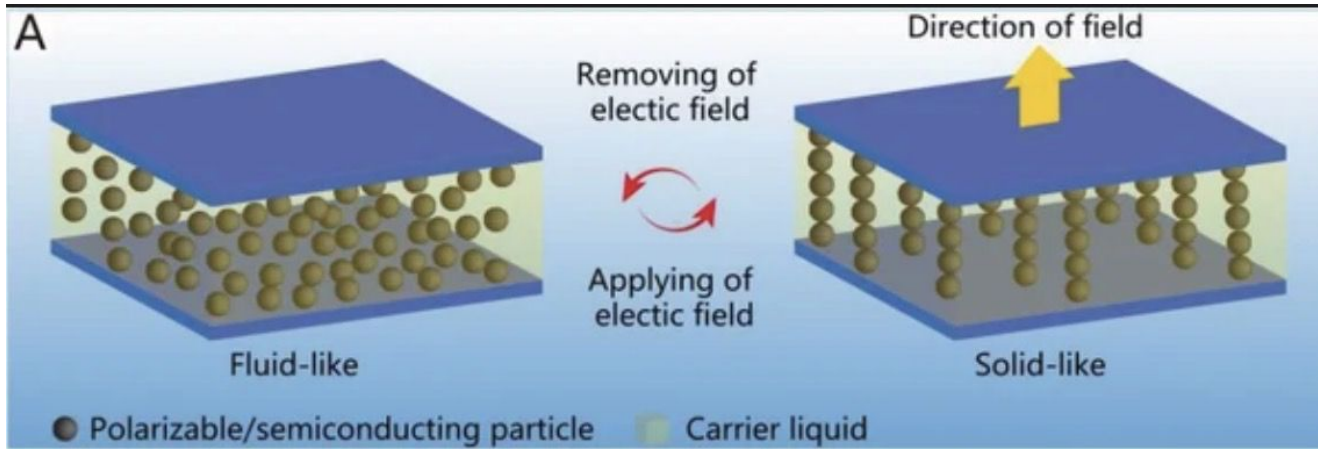
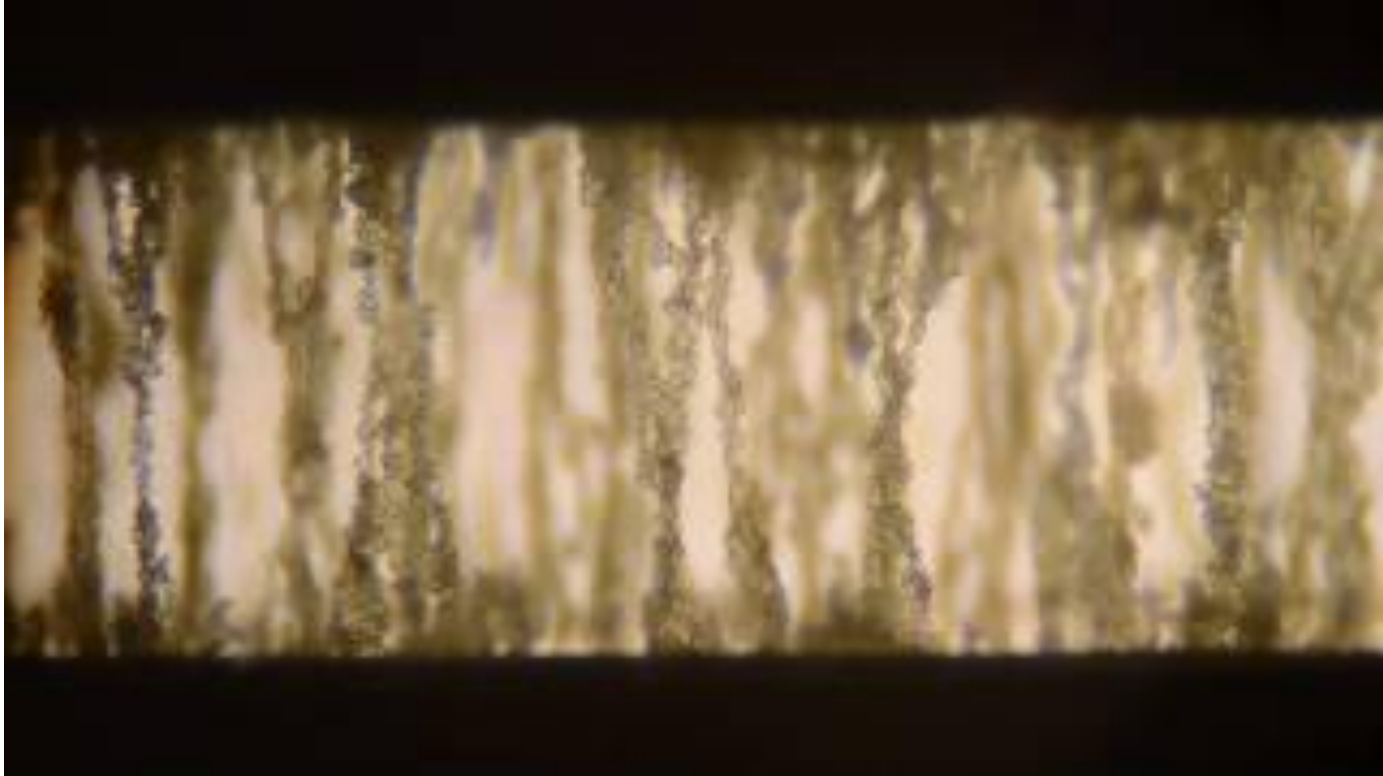


Theory

Applied electric field induces dipole in suspended particles, which align into chains



Microscopy



Theory

Yield Stress Electric Field Relation

$$\tau_y = \alpha E^2$$

Analytical Polarization Coefficient

$$\alpha = 3\phi\epsilon_0\epsilon_r\beta^2$$

β - Clausius-Mossotti factor

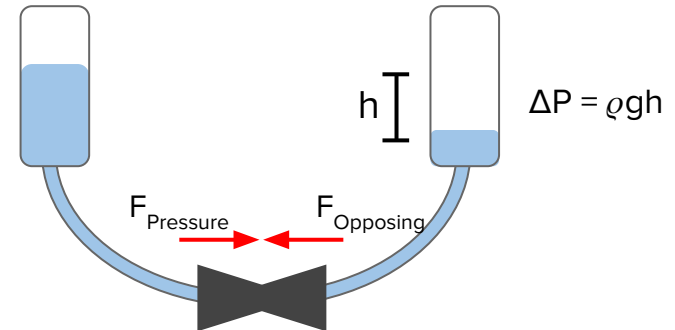
ϕ - volume fraction

ϵ - permittivity

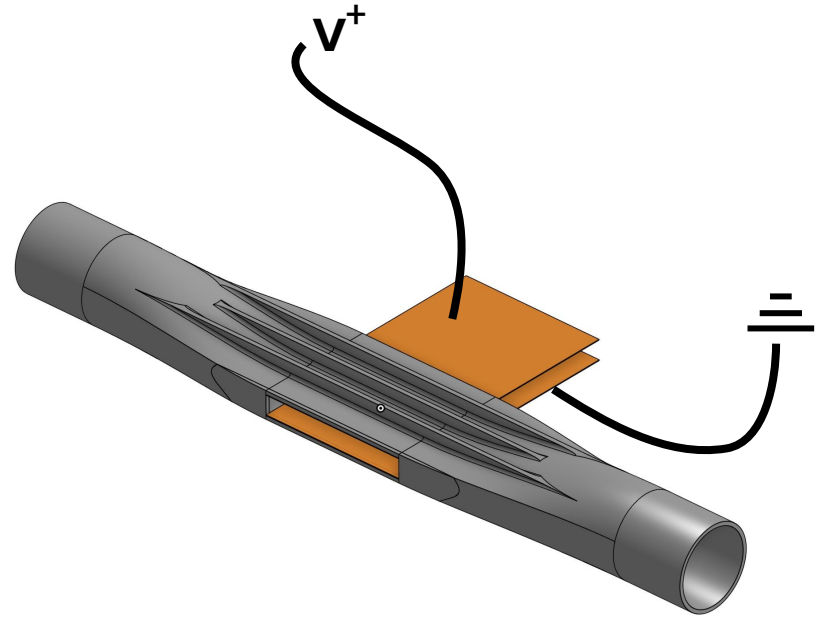
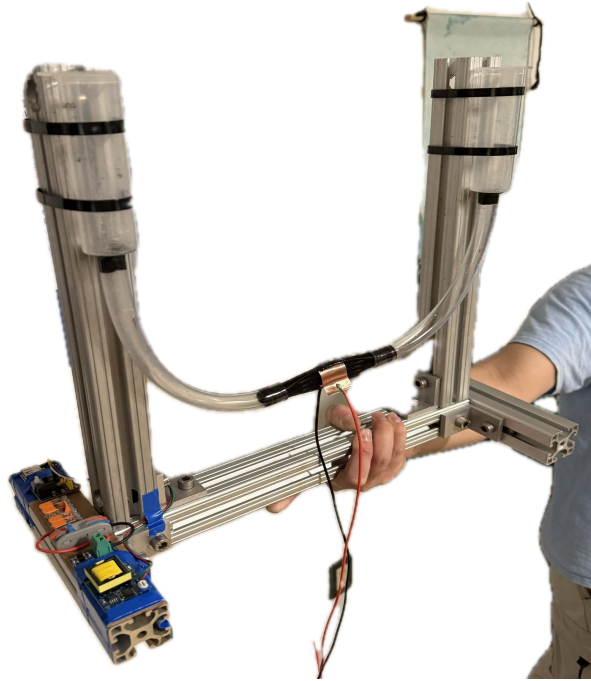
Experimental Test

$$F_{Pressure} = F_{opposing}$$
$$\Delta P \cdot w \cdot d = \tau_y \cdot (2wl)$$

$$\tau_y = \frac{\Delta P d}{2l}$$



Setup

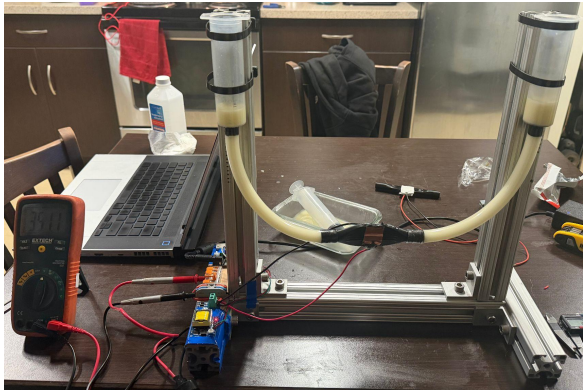


Gap (d) = 3.1mm

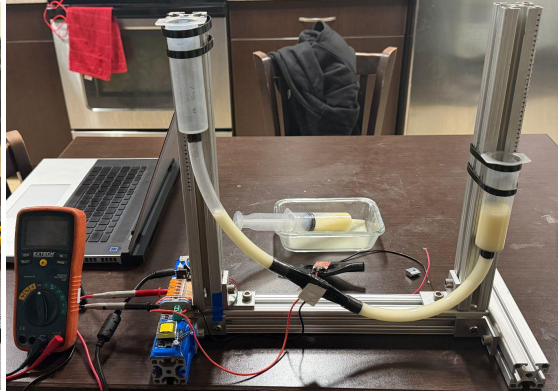
Length (l) = 25mm

Peak Field Intensity ($|E|$) @ 1050 Volts = 338709.6774 V/m

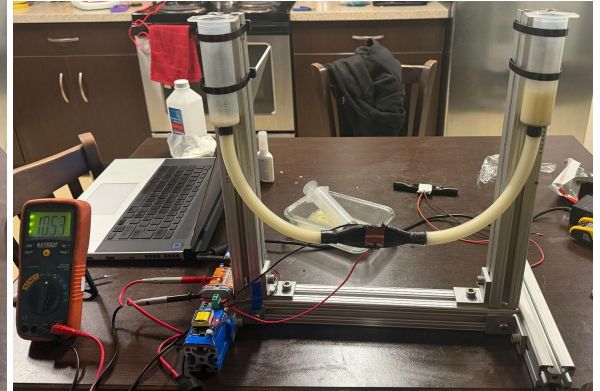
Experiment



Step 1:
Equal Pressure



Step 2:
Create Height
Difference



Step 3:
Energize and
Reset Height

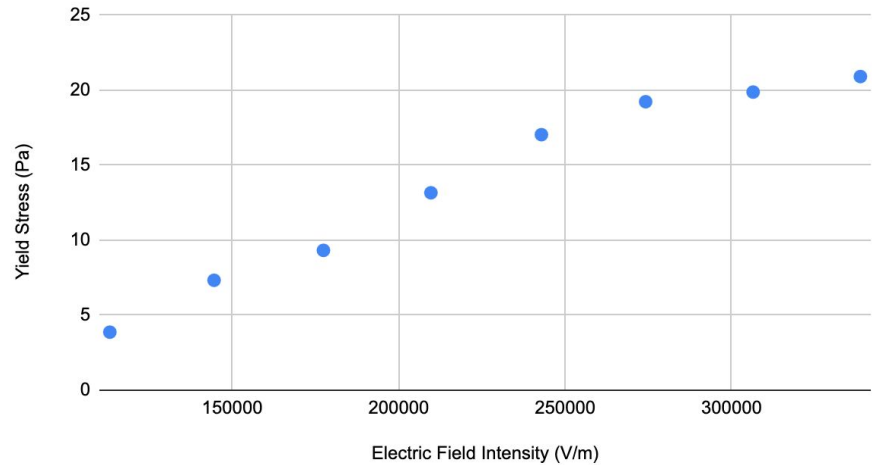
Results

Voltage (V)	Height Difference (m)	Yield Stress(Pa)
0	0	0
351	0.0060	3.87
448	0.0113	7.33
550	0.0144	9.31
650	0.0204	13.16
753	0.0264	17.03
850	0.0298	19.23
950	0.0308	19.87
1050	0.0324	20.91

$$\tau_y = \alpha E^2$$

$$\alpha = 1.87\text{E-}10 \frac{\text{Pa}\cdot\text{m}^2}{\text{V}^2}$$

Yield Stress vs Electric Field Intensity



MAE/ECE 535: DESIGN OF ELECTROMECHANICAL SYSTEMS

Experimental Demonstration

Electrorheological Fluid Rheometer

Jack Fowler, Vincent Wu

March 11, 2026

Abstract

In this experiment, the electrorheological (ER) effect was observed by using an ER fluid made of cornstarch and canola oil. To find the electromechanical relationship between the yield stress of the ER fluid and the electric field applied to the ER fluid, a rheometer was constructed. This rheometer utilized two reservoirs coupled together and set at different heights to generate a pressure head. The coupling passes the ER fluid through an electric field. The yield stress τ_y at various electric field strengths could then be found by applying a known pressure to the fluid. This yielded a polarization coefficient α of $1.87\text{E-}10 \frac{\text{Pa}\cdot\text{m}^2}{\text{V}^2}$ that relates the yield strength to the electric field.

Introduction

Electrorheological (ER) fluids are characterized by the ability to transition from a liquid state to a near-solid state when subjected to an external electric field. These fluids are made by mixing polarizable particles in a non-conductive carrier. In the absence of an electric field, the fluid behaves like a Newtonian fluid. However, when subjected to a sufficient electric field, the particles are polarized and form chain-like structures that inhibit the flow of the fluid.

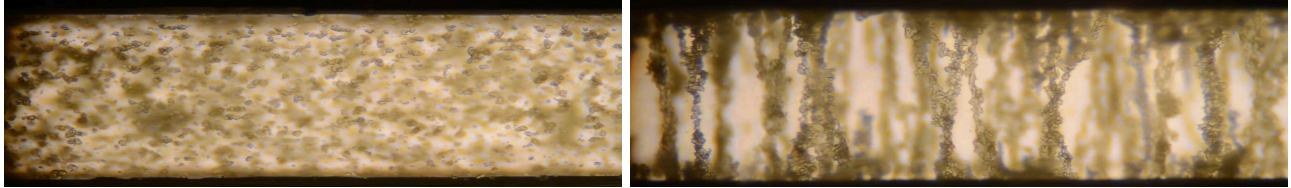


Figure 1) Video of particles being polarized and chaining ^[3]

The transition from a Newtonian fluid to a semi-solid is described using the Bingham Plastic model. In this model, the fluid remains rigid until the applied shear stress exceeds the yield stress τ_y . Once this threshold is surpassed, the material flows as a viscous fluid. The yield stress of an ER fluid can be related to the electric field it was subjected to with a polarization coefficient (α), derived from a point-dipole polarization model.^[2]

$$\tau_y = \alpha E^2$$

The polarization coefficient is defined as follows, where β is the Clausius-Mossotti factor (the relative polarizability of the particles compared to the continuous fluid), ϕ is the volume fraction, and ϵ is the permittivity.

$$\alpha = 3\phi\epsilon_0\epsilon_r\beta^2$$

To find the polarization coefficient of an ER fluid made from a mixture of cornstarch and canola oil, a rudimentary hydrostatic pressure-based rheometer was constructed.

Methods and Materials

Fabrication

An ER fluid was created by mixing regular cornstarch and canola oil at a volume fraction ($\phi \approx 25\%$), cited as optimal for low unenergized viscosity with a high field-induced yield stress^[1]. The solution was well mixed, then the density was measured with a 60ML syringe and scale ($\rho = 1061.16 \text{ kg/m}^3$).

Electrical components were ordered to apply the electric field across two thin aluminum plate electrodes. A 1000V maximum voltage was chosen, based on boost converter availability and measurement equipment maximums. The boost converter has a potentiometer to adjust the output voltage from 350V to 1000V.

A test stand was fabricated using 80-20 aluminum extrusion, PVC tubing, syringes for reservoirs, and a restriction valve was created out of PLA using FDM 3D printing, with print settings for a watertight enclosure. The electrode gap was sized such that an electric field magnitude sufficient to polarize could be achieved.^[1] The electrodes were then assembled into the 3D printed part and soldered to the high-voltage output (with an inline multimeter). The entire assembly of the electrode body, tubes, and reservoirs was sealed using JB-Weld 50139. The fluid can flow from one reservoir to another, passing through the restrictor body and electric field.

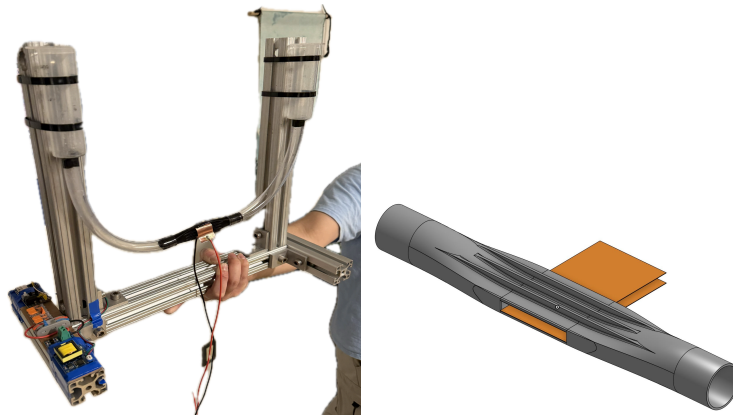


Figure 2) U-bend Rheometer Design

Experiment

The system was filled with the ER fluid and purged of air. The static equilibrium fluid level was measured with no applied voltage, then measurements at increasing voltages began. First, the right reservoir was moved down to change the fluid height. Second, voltage was set using the potentiometer on the board, and the electrodes were energized. Third, the right reservoir was moved back to the original height smoothly. 45 seconds were allowed to pass for the system to reach equilibrium, and then the height difference of the fluid was measured. This was repeated from 350V to 1050V.

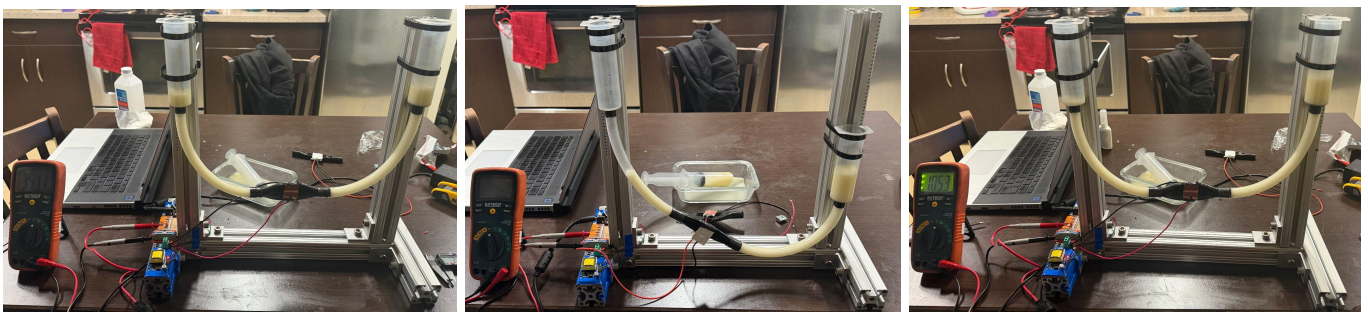


Figure 3) Measurement process

Results

Using the height difference data collected, the hydrostatic pressure applied to the solidified electrorheological fluid was calculated, and equated to the internal shear of the liquid. This assumed a no-slip boundary condition and that the shear

grip on the 3mm sides was negligible. The length l was 25mm, and the gap d (after accounting for electrode thickness) was 3.1mm.

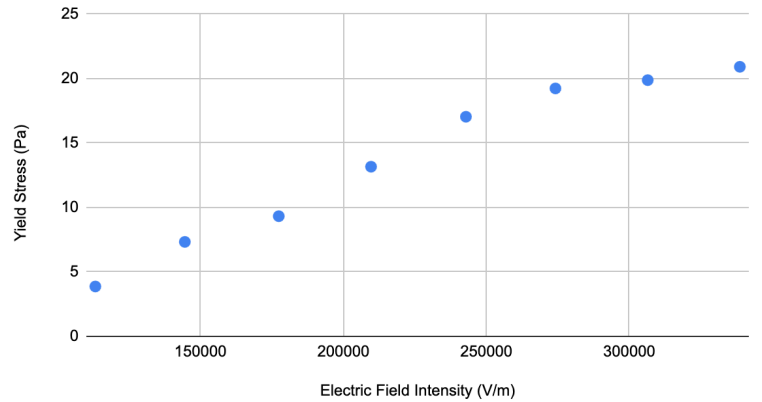
$$F_{Pressure} = F_{opposing}$$

$$\Delta P \cdot w \cdot d = \tau_y \cdot (2wl)$$

$$\tau_y = \frac{\Delta P d}{2l}$$

Voltage (V)	Height Difference (m)	Yield Stress(Pa)
0	0	0
351	0.006	3.87253674
448	0.01136	7.332002894
550	0.01444	9.319905088
650	0.0204	13.16662492
753	0.0264	17.03916166
850	0.0298	19.23359914
950	0.0308	19.87902193
1050	0.0324	20.9116984

Yield Stress vs Electric Field Intensity



Using this data and fitting yield stress to the square of the electric field, our experimental polarization coefficient comes out to $1.87\text{E-}10 \frac{\text{Pa}\cdot\text{m}^2}{\text{V}^2}$. This can be compared to the analytically calculated value below, with our volume fraction, canola oil's permittivity^[4] of 3.1, and an assumed maximum theoretical dielectric limit for the Clausius-Mossotti factor ($\beta = 1$).

$$\alpha = 3 \cdot 0.25 \cdot 8.854 \cdot 10^{-12} \cdot 3.1 \cdot 1^2 = 2.06 \cdot 10^{-11} \frac{\text{Pa}\cdot\text{m}^2}{\text{V}^2}$$

Additionally, this value can be compared to existing literature on the electrorheological behavior of some native starches in ER fluids, with researchers finding similar yield stresses and α values of $3.75\text{E-}11 \frac{\text{Pa}\cdot\text{m}^2}{\text{V}^2}$.^[1]

Conclusions

This experiment successfully demonstrated the electromechanical behavior of an ER fluid made from cornstarch and canola oil. The data gathered from the homemade rheometer allowed the yield stress and electric field relationship of the ER fluid to be characterized. By varying the electric field strength applied to the ER fluid, the yield stress of the fluid could be controlled. This gave a clear relationship between the yield stress and the electric field represented by the polarization coefficient α . Our experimentally found values align with those found by researchers in literature, with variation attributed to the humidity of the polarized molecules, with ours being a wet-base, and researchers commonly using fully desiccated dry-base particles. The analytically calculated value for polarization coefficient assumes an ideal polarization model and does not reflect many of the real-world interactions.

References

- [1] Yavuz, Mustafa, et al. "Electrorheological Behavior of Biodegradable Modified Corn Starch/Corn Oil Suspensions." *Carbohydrate Polymers*, vol. 79, no. 2, Jan. 2010, pp. 318–324, <https://doi.org/10.1016/j.carbpol.2009.08.008>. Accessed 30 May 2020.
- [2] Seo, Yongsok. "A New Yield Stress Scaling Function for Electrorheological Fluids." *Journal of Non-Newtonian Fluid Mechanics*, vol. 166, 2011, pp. 241–243, <https://doi.org/10.1016/j.jnnfm.2010.11.010>. Accessed 12 Mar. 2026.
- [3] Semenov, Nikolay. "Electrorheology." YouTube, 13 Oct. 2014, <https://www.youtube.com/watch?v=ybyeMw1b0L4>. Accessed 12 Mar. 2026.
- [4] Peñaloza-Delgado, R., et al. "Dielectric Characterization of Vegetable Oils During a Heating Cycle." *Journal of Food Science and Technology*, vol. 58, no. 4, 2021, pp. 1480–1487, <https://doi.org/10.1007/s13197-020-04660-7>. Accessed 12 Mar. 2026.